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Spin Correlation in the Ground State of Spin-1 Bosons Beyond the Single-Mode Approximation

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June 05, 2026

OUTLINE

1. Background
2. Model
3. Methodology
4. Results
5. Conclusion

Hyperfine spin of an atom

$$F = |S + L \pm I|$$

- S: the electron spin
- L: the electron orbital angular momentum
- I: the nuclear spin

Internal degrees of freedom of Bose-Einstein condensates

- In magnetic trap: spin degrees of freedom are frozen.
- In optical trap: spin degrees of freedom are liberated.
  → **Spinor Bose-Einstein condensate**

Spin-1 Bose-Einstein condensate $(F = 1)$

$$\mathbf{\Psi} = \begin{pmatrix} \psi_1 \\ \psi_0 \\ \psi_{-1} \end{pmatrix}$$

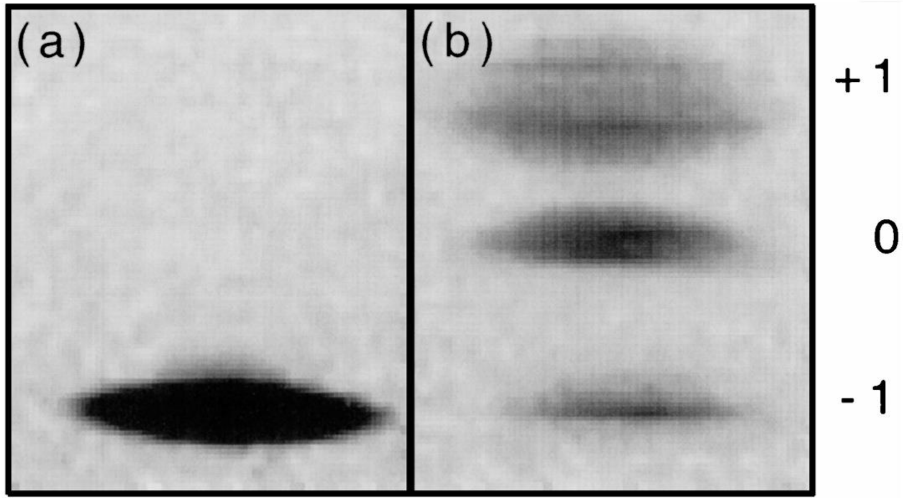
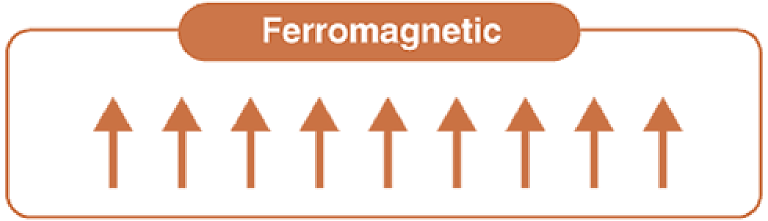


Fig. 1. Absorbtion images of optical trapping of condensates (^{23}Na) in all $F = 1$ hyperfine states. [1]
(a) Atoms remain spin polarized in the optical trap.
(b) Atoms populated all hyperfine states (by using a radio frequency sweep).

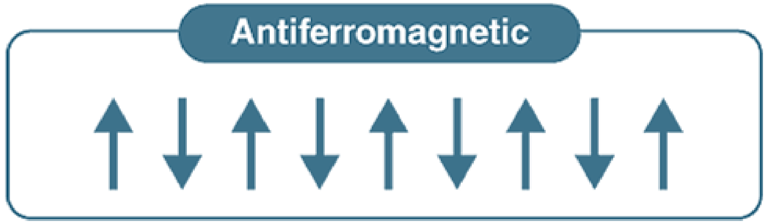
[1] D.M. Stamper-Kurn *et al.*, *Phys. Rev. Lett.* **80**, 2027 (1998)

Mean-field

^{87}Rb



^{23}Na



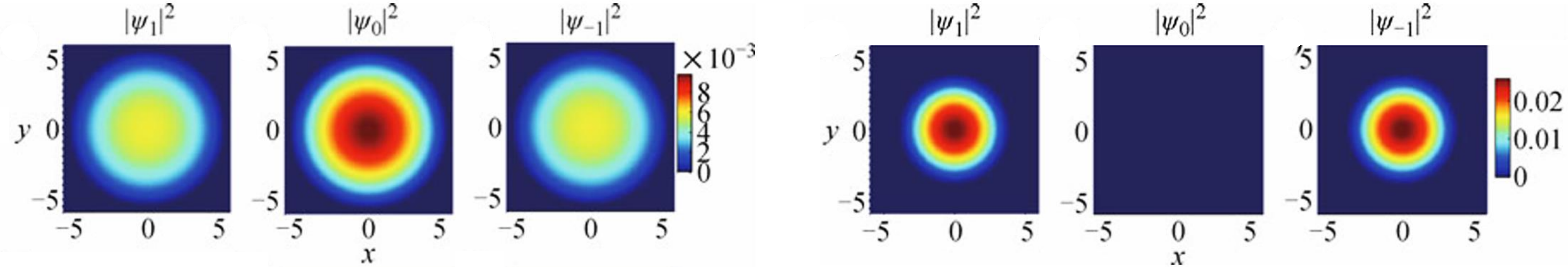


Fig 2. The three components density distributions for spinor ^{87}Rb and ^{23}Na condensates [2].

[2] S.-W. Song et. al., “Ground states, solitons and spin textures in spin-1 Bose-Einstein condensates,” *Frontiers of Physics*, 2013.

Many-body

Small BEC of spin-1 atoms is tightly confined → Single-mode approximation (SMA)

Anti-ferromagetic interaction

Adiabatically ramped down the external magnetic field to near zero

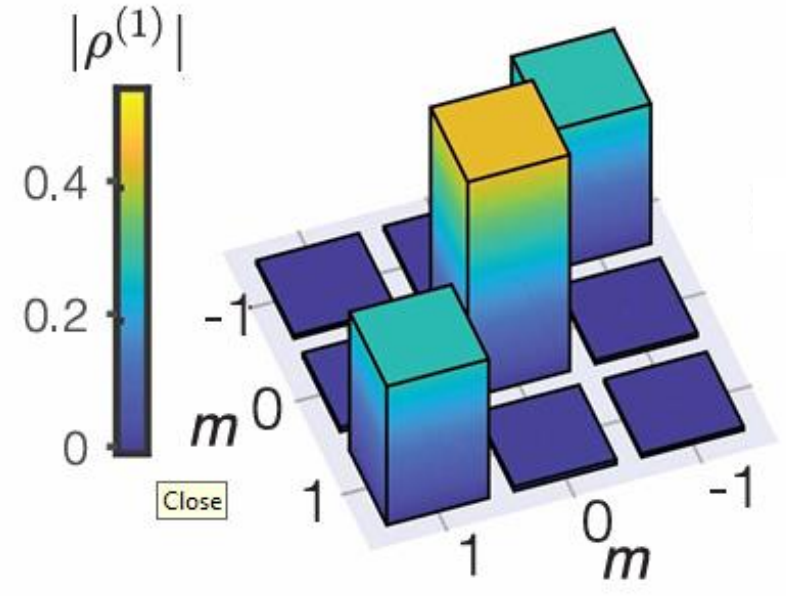


Fig. 4. The moduli of the elements of one-body density matrix for fragmented state [3]

[3] B. Evrard et. al., "Observation of fragmentation of a spinor Bose-Einstein condensate," *Science*, 2021.

2. MODEL

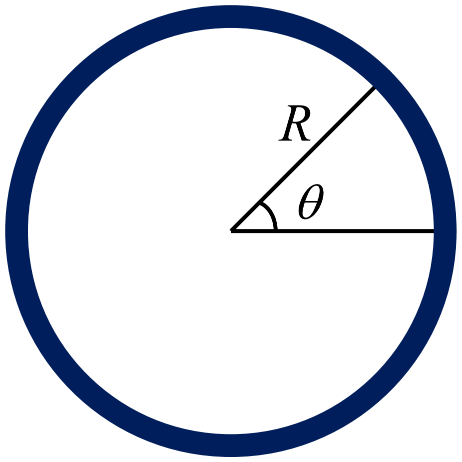


Fig. 3: Schematic illustration of a quasi-1D torus.

$$\hat{H} = \hat{K} + \hat{H}_0 + \hat{H}_1$$

$$= \int_0^{2\pi} d\theta \left(\underbrace{-\hat{\psi}_m^\dagger \frac{\partial^2}{\partial\theta^2}\hat{\psi}_m}_{\text{Kinetic}} + \underbrace{g_0 \hat{\psi}_{m_1}^\dagger \hat{\psi}_{m_2}^\dagger \hat{\psi}_{m_2} \hat{\psi}_{m_1}}_{\text{Spin-independent interaction}} + \underbrace{g_1 \hat{\psi}_{m_1}^\dagger \hat{\psi}_{m_1'}^\dagger \mathbf{f}_{m_1 m_2} \cdot \mathbf{f}_{m_1' m_2'} \hat{\psi}_{m_2'} \hat{\psi}_{m_2}}_{\text{Spin-dependent interaction}} \right)$$

$g_0 < 0$: attractive

$g_0 > 0$: repulsive

$g_1 < 0$: ferromagnetic

$g_1 > 0$: polar

The field operator: $\hat{\psi}_m(\theta) = \boxed{\sum_l} \frac{1}{\sqrt{2\pi}} \hat{a}_m^l e^{il\theta}$ where $\hat{a}_m^l$ is the annihilation operator of a boson with magnetic quantum number m and angular momentum l.

Single-mode approximation (SMA)

$l = 0$

Beyond SMA (This study)

$l = -m_l, \ldots, m_l$

3. METHODOLOGY

EXACT DIAGONALIZATION

Hilbert space → Hamiltonian matrix → Diagonalization → Eigen energies /states

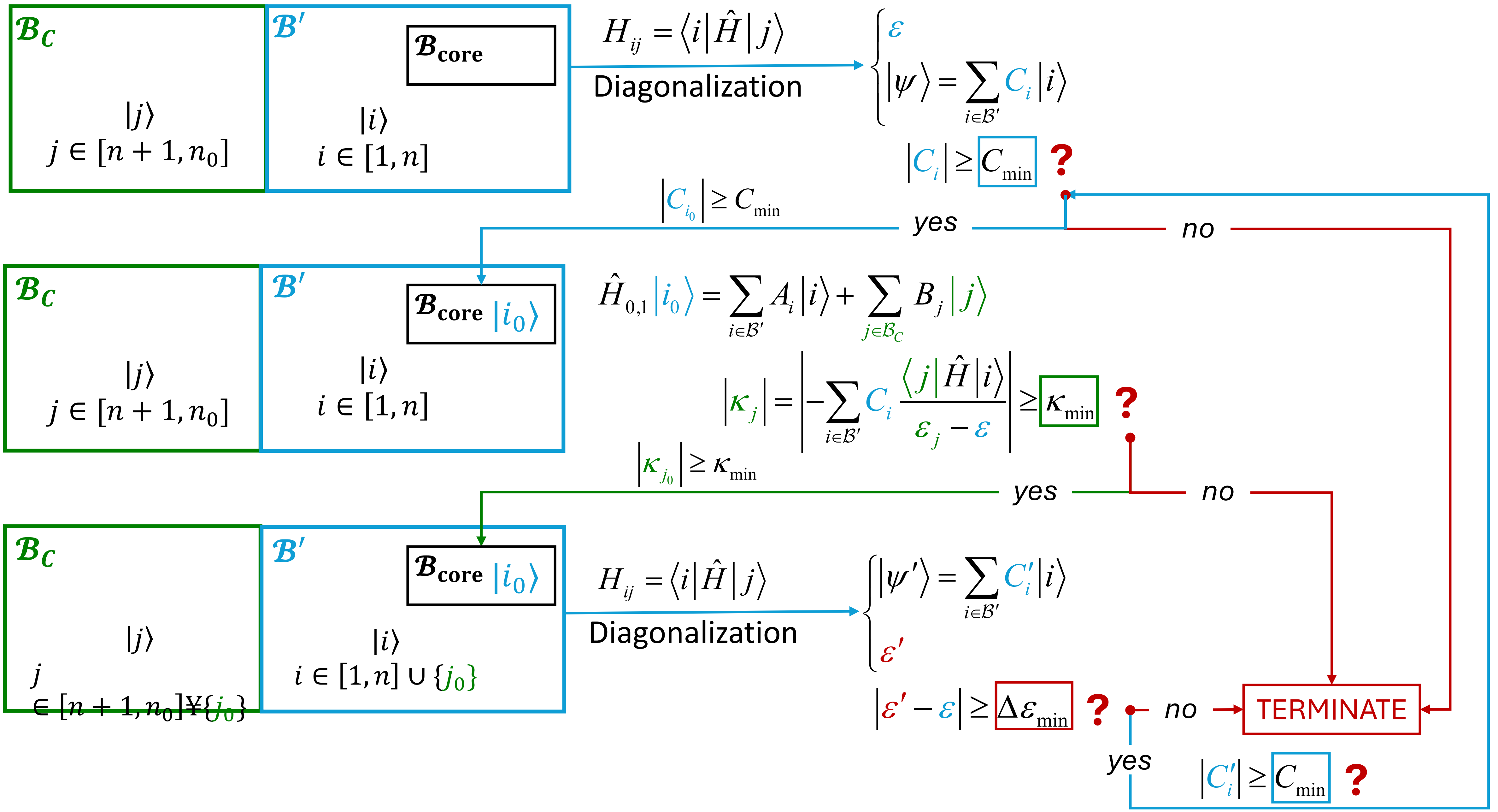
1. One-body basis truncation:

One-body basis: $\left|\phi_{m,l}\right\rangle$ with $-m_l \le l \le m_l$, $\boxed{m_l \le m_{l_{\max}}}$

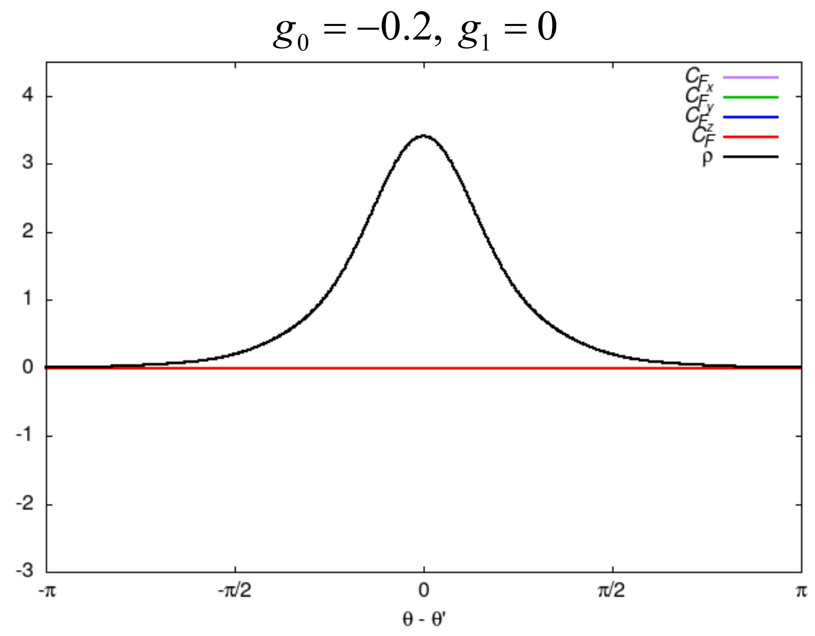
2. Importance Truncation Configuration Interaction (ITCI):

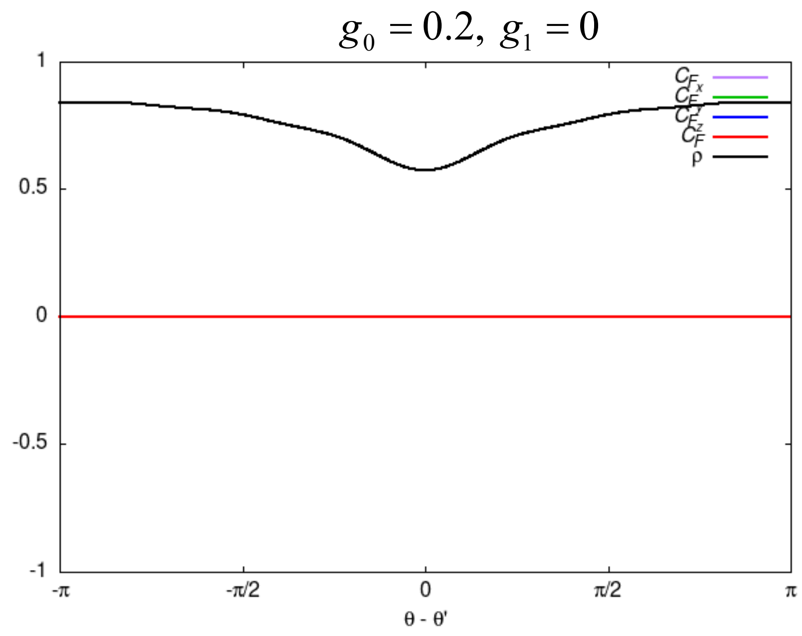
Retaining only the configurations with the most significant contributions to the wavefunction.

Importance Truncation Configuration Interaction (ITCI)



No spin interaction



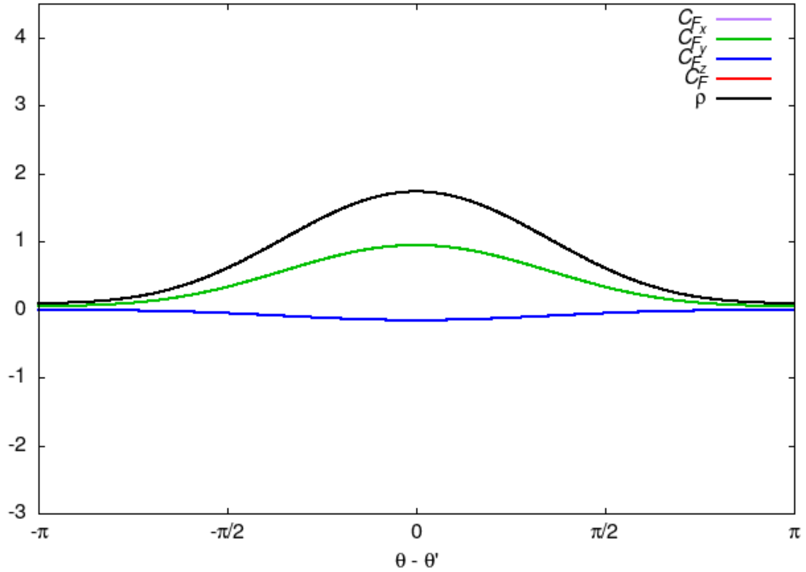


Spatial bunching **Spatial anti-bunching**

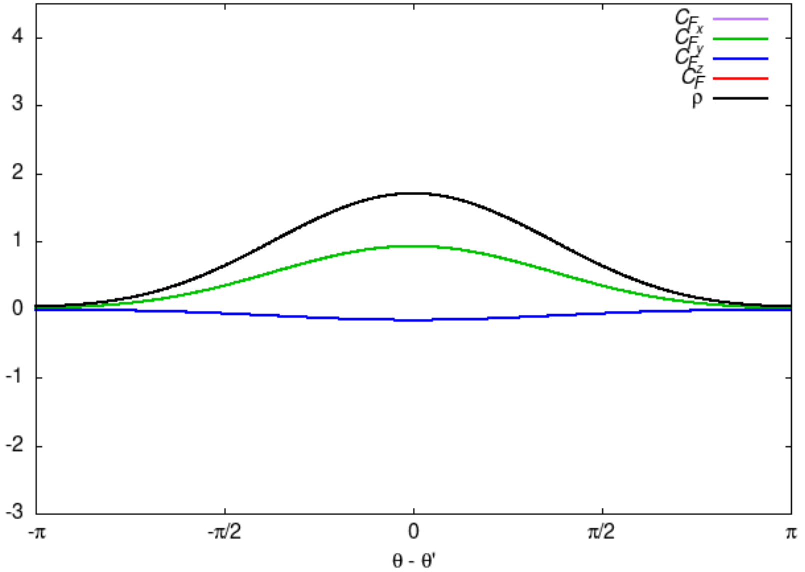
All spin remain exactly zero

Ferromagnetic interaction

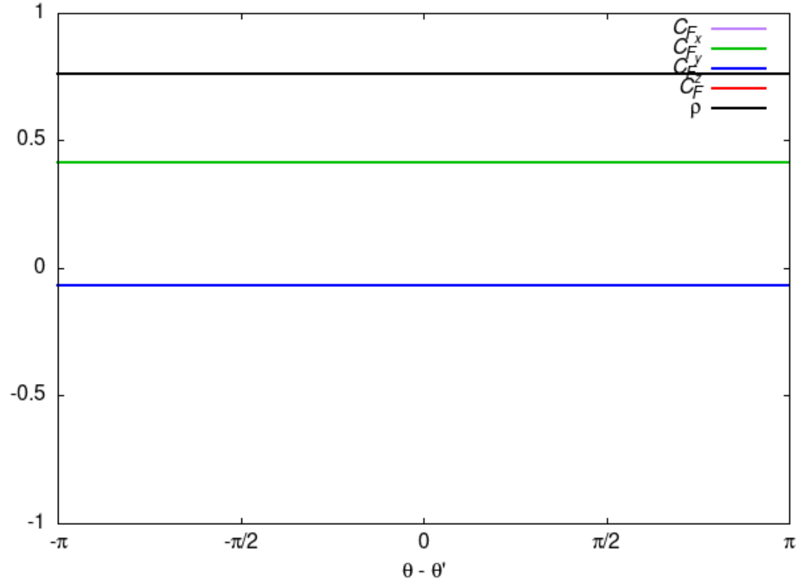
$g_0 \approx 0,\ g_1 = -0.2$



$g_0 = -0.2,\ g_1 = -0.2$



$g_0 = 0.2,\ g_1 = -0.2$



Spatial bunching and spin-correlated.

Nearby atoms tend to align their transverse spin components, anti-align of the longitudinal spin component.

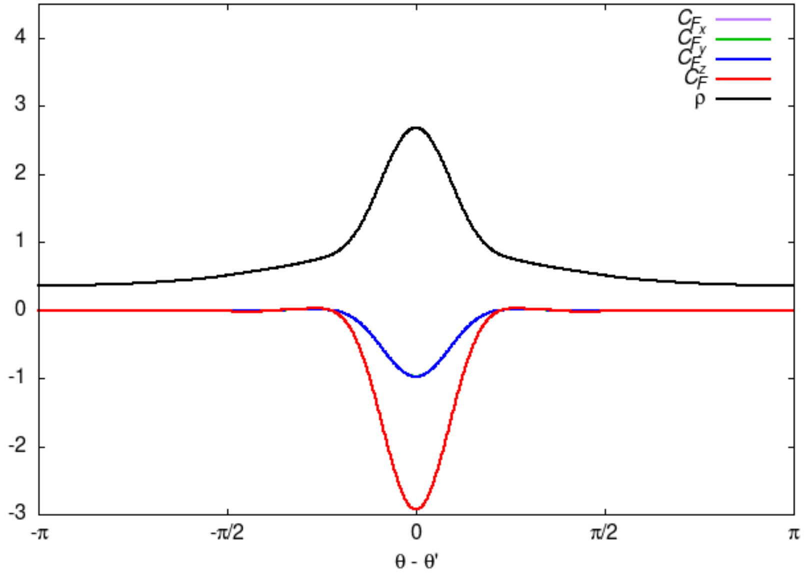
Homogeneous particle distribution

Spatially uniform spin correlations

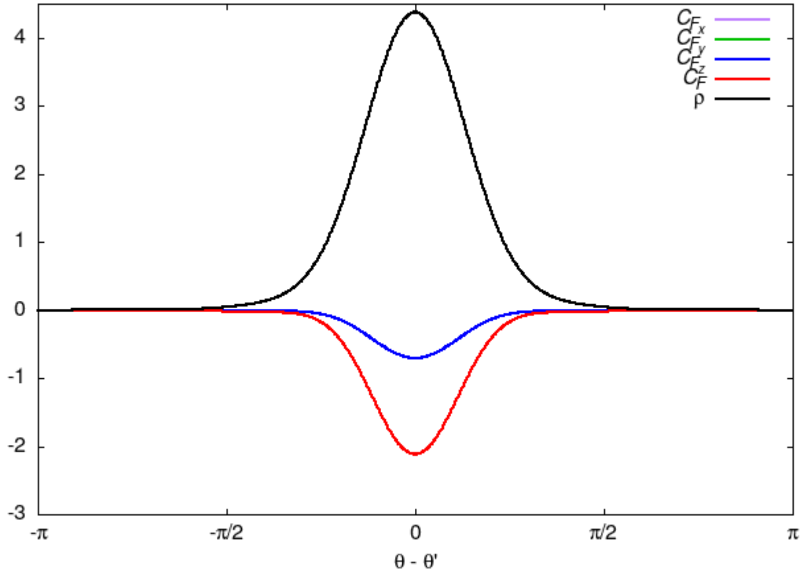
All atoms exhibit the same correlation regardless of distances: transverse spin components tend to align, longitudinal spin components tend to anti-align.

Anti-ferromagnetic interaction

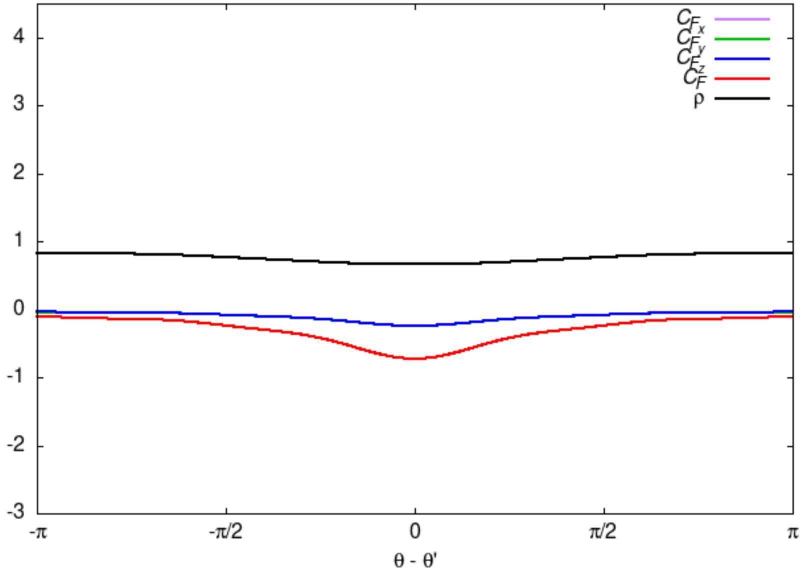
$g_0 \approx 0,\ g_1 = 0.2$



$g_0 = -0.2,\ g_1 = 0.2$



$g_0 = 0.2,\ g_1 = 0.2$



Strong spatial bunching

Weak spatial anti-bunching

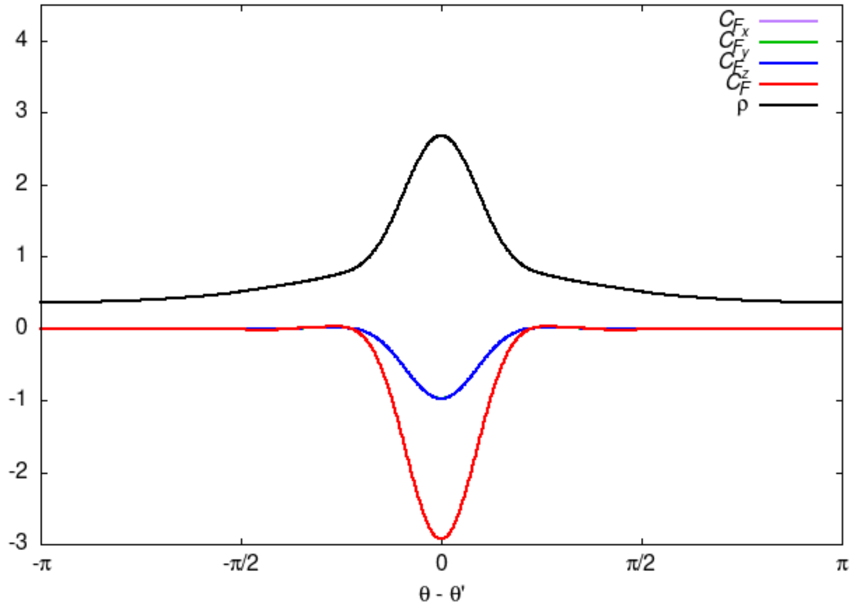
Spin anti-correlated.

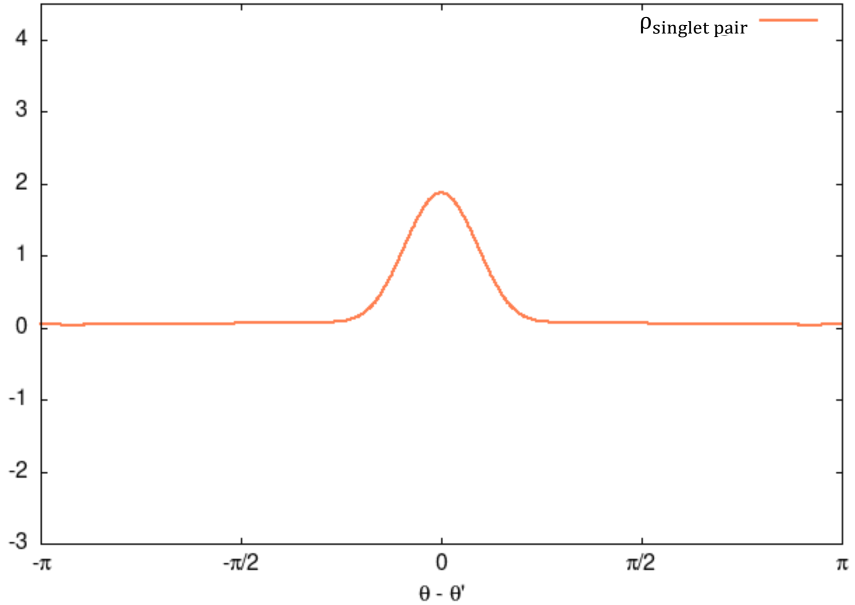
Nearby atoms tend to have anti-correlated spins in all directions.

Spin correlation remains isotropic.

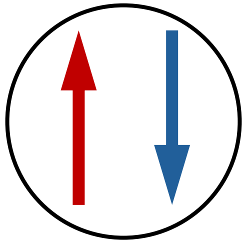
4. RESULTS

$g_0 \approx 0,\ g_1 = 0.2$

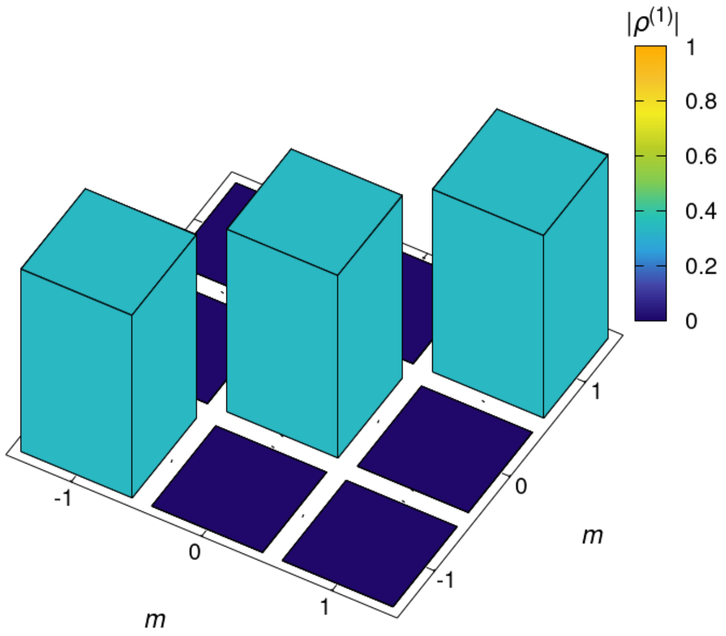




Spin-singlet pair



Total collective spin $S = 0$



Consistent with SMA

5. CONCLUSION

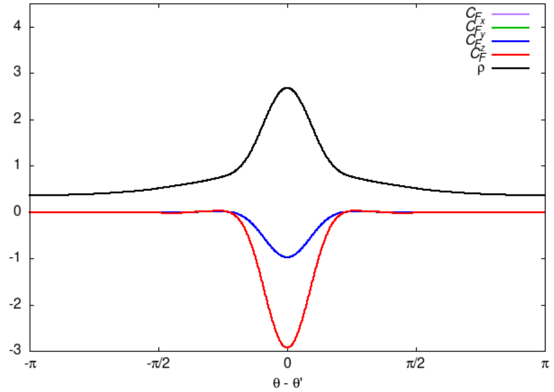
SUMMARY

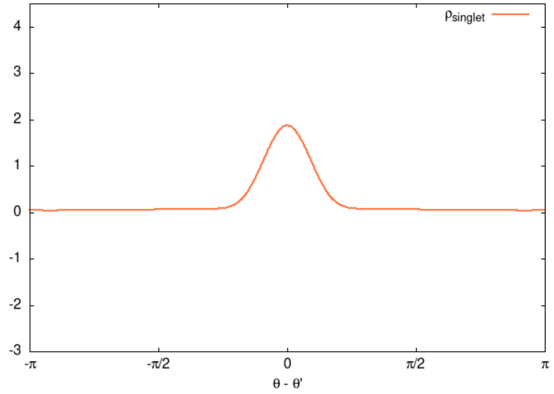
Ferromagnetic interaction $g_1 < 0$

- The system minimizes its energy by aligning all spins parallel to each other, thereby maximizing the total collective spin.
- $g_0 < 0$: Spin spatial bunching alignment.
- Strong $g_0 > 0$: Spatially uniform spin alignment

Anti-ferromagnetic interaction $g_1 > 0$

- Spin correlations are highly isotropic and strongly anti-correlated.
- The system minimizes energy by forming a fragmented macroscopic spin-singlet state.
- Atoms spontaneously cluster to form localized spin-singlet pairs.





FURTHER PROSPECTS

- Examine spin-singlet soliton formation.
- Investigate properties of low-lying excited states.
- Use other methodologies to increase number of atoms.